

#

2082 Bh

$$h(n) = 1, 2, 1, -1 \quad x(n) = 1, 2, 2, 1$$

Here,

$$y[m] = \sum_{n=0}^{N-1} x[n] h[m-n]_N$$

$$= \sum_{n=0}^3 x[n] h[m-n]_4$$

$$= x[0] h[m]_4 + x[1] h[m-1]_4 + x[2] h[m-2]_4 + x[3] h[m-3]_4$$

$$= h[m]_4 + 2h[m-1]_4 + 3h[m-2]_4 + h[m-3]_4$$

$$y[0] = h[0]_4 + 2h[3]_4 + 3h[2]_4 + h[1]_4$$

$$= 1 + 2(-1) + 3(2) + 2$$

$$= 4$$

$$y[1] = h[1]_4 + 2h[0]_4 + 3h[3]_4 + h[2]_4$$

$$= 2 + 2 + 3(-1) + 1$$

$$= 2$$

$$y[2] = h[2]_4 + 2h[1]_4 + 3h[0]_4 + h[3]_4$$

$$= 2 + 4 + 3 + (-1)$$

$$= 7$$

$$y[3] = h[3]_4 + 2h[2]_4 + 3h[1]_4 + h[0]_4$$

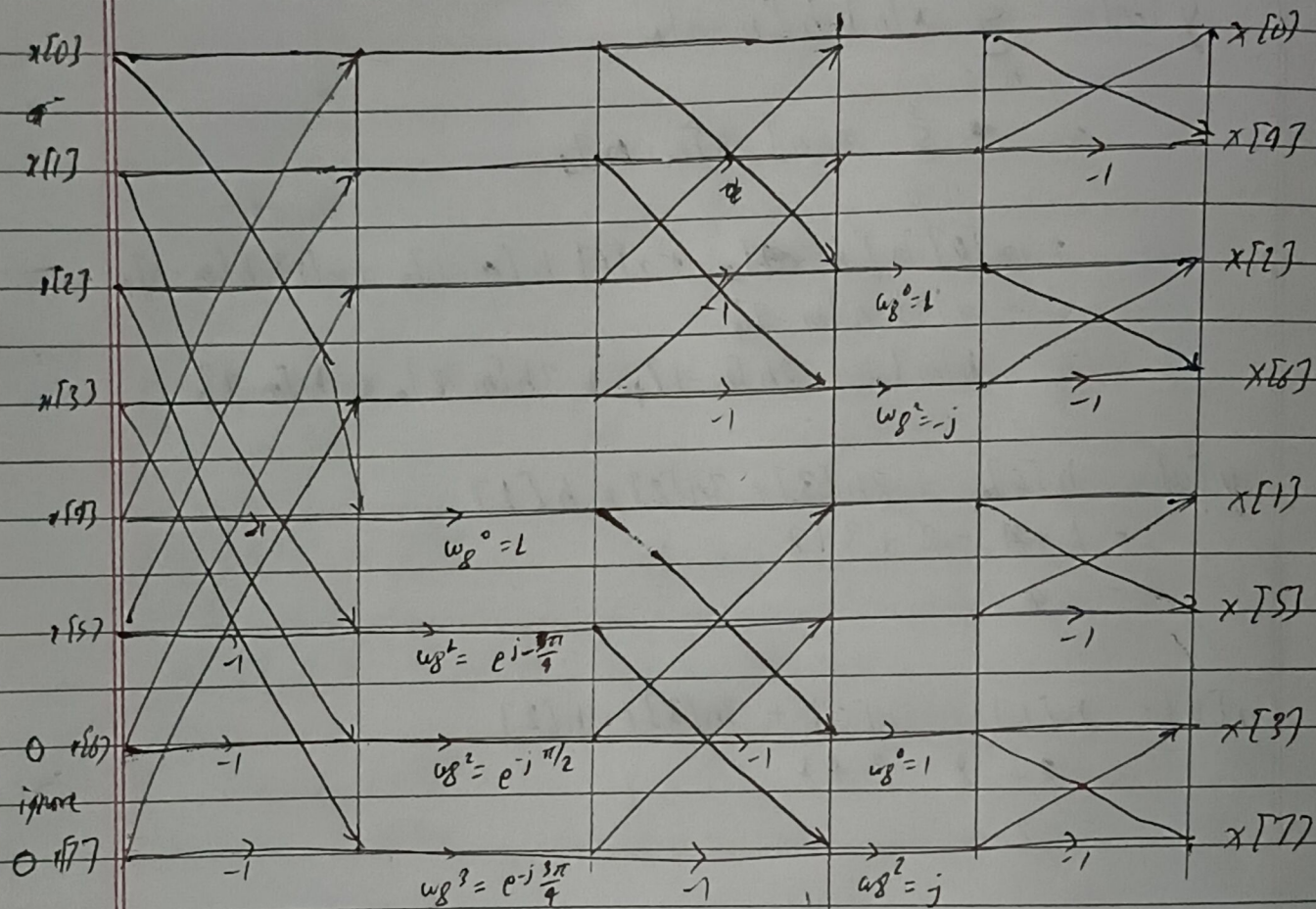
$$= (-1) + 2(2) + 3(2) + 1$$

$$= 4$$

DIF-FFT

$$x[n] = \{1, \frac{1}{2}, -1, -\frac{1}{2}, 2, -\frac{3}{2}\}$$

add 2 more 0 as bit-zero padding



$$X[0] = x[0] + x[1] + x[2] + x[3] + x[4] + x[5] + x[6] + x[7]$$

$$= \frac{1}{2}$$

$$X[4] = x[0] - x[1] + x[2] - x[3] + x[4] - x[5] + x[6] - x[7]$$

$$= \frac{7}{2}$$

$$X[2] = x[0] + w_8^2 x[1] - x[2] - w_8^2 x[3] + x[4] + w_8^2 x[5]$$

$$= 4 + j\frac{1}{2}$$

$$X[3] = x[0] + w_8^1 x[1] + w_8^2 x[2] + w_8^3 x[3] - x[4] - w_8^1 x[5] - w_8^2 x[6] - w_8^3 x[7]$$

$$= 0.767 - j0.06$$

$$X[5] = x[0] - \omega_8^1 x[1] + \omega_8^2 x[2] - \omega_8^3 x[3] + x[4] + \omega_8^1 x[5]$$
$$= -2.767 + j2.06$$

So:

$$X[0] = \frac{1}{2}$$

$$X[1] = 0.767 - j0.06$$

$$X[2] = 4 + j0.5$$

$$X[3] = X[5]^* = -2.767 - j2.06$$

$$X[4] = 3.5$$

$$X[5] = -2.767 + j2.06$$

$$X[6] = X[2]^* = 4 - j0.5$$

$$X[7] = X[1]^* = 0.767 + j0.06$$

82 BA

Fast Fourier Transform is an efficient algorithm that computes Discrete Fourier Transform (DFT) or its inverse.

in DFT:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j \frac{2\pi kn}{N}}$$

multiplication: N^2 multiplications betⁿ $x[n]$ and $e^{j \frac{2\pi kn}{N}}$
and again $N(N-1)$ additions, within 1 sum for N terms.

had, slow, high computation.

in FFT:

$$X[k] = \sum_{n=0}^{N-1} x[n] \omega_N^{kn}$$

↑
twiddle factor

split $x[n]$ into odd, even,

$$X[k] = \sum_{m=0}^{\frac{N}{2}-1} x[2m] \omega_N^{2km} + \sum_{m=0}^{\frac{N}{2}-1} x[2m+1] \omega_N^{2k(m+1)}$$

$$= \sum_{m=0}^{\frac{N}{2}-1} x[2m] \omega_N^{2km} + \omega_N^k \sum_{m=0}^{\frac{N}{2}-1} x[2m+1] \omega_N^{2km}$$

↓
DFT

↓
DFT

so, since we are recursively dividing each DFT into other DFT's, there is less multiplication/addition in each step. later we can simply combine these split DFT's into one final FFT.

DFT $\rightarrow n^2$

slow

FFT: $\rightarrow \frac{N}{2} \log_2 N$

faster

if asked quantitatively elaborate / table

on $O(n)$ for these 2, in 2, 4, 8,

to show FFT very less computation

$$N = 4$$

2x padding by 1 bit
01F-FF7



$x[0]$

$x[1]$

$x[2]$

$x[3]$

$x[0]$

$x[1]$

$x[2]$

$x[3]$

$x[0]$

$x[1]$

$x[2]$

$x[3]$

$x[0]$

$x[1]$

$x[2]$

$x[3]$

$x[0]$

$x[1]$

$x[2]$

$x[3]$

$$X[0] = x[0] + x[1] + x[2] + x[3] = 4$$

$$X[1] = x[0] - jx[1] - x[2] + jx[3] = 2\sqrt{2}j$$

$$X[2] = x[0] + -jx[1] + x[2] = -4$$

$$X[3] = x[0] + x[1] - x[2] + x[3] = 1 + j4$$

81 an

$$x_1[n] = \{1, 1, -1, -1, 1\} \rightarrow 8 \text{ bits}$$

$$x_2[n] = \{1, -1, -1\} \rightarrow 8 \text{ bits}$$

$$y[m] = \sum_{n=0}^7 x_1[n] x_2[m-n]$$

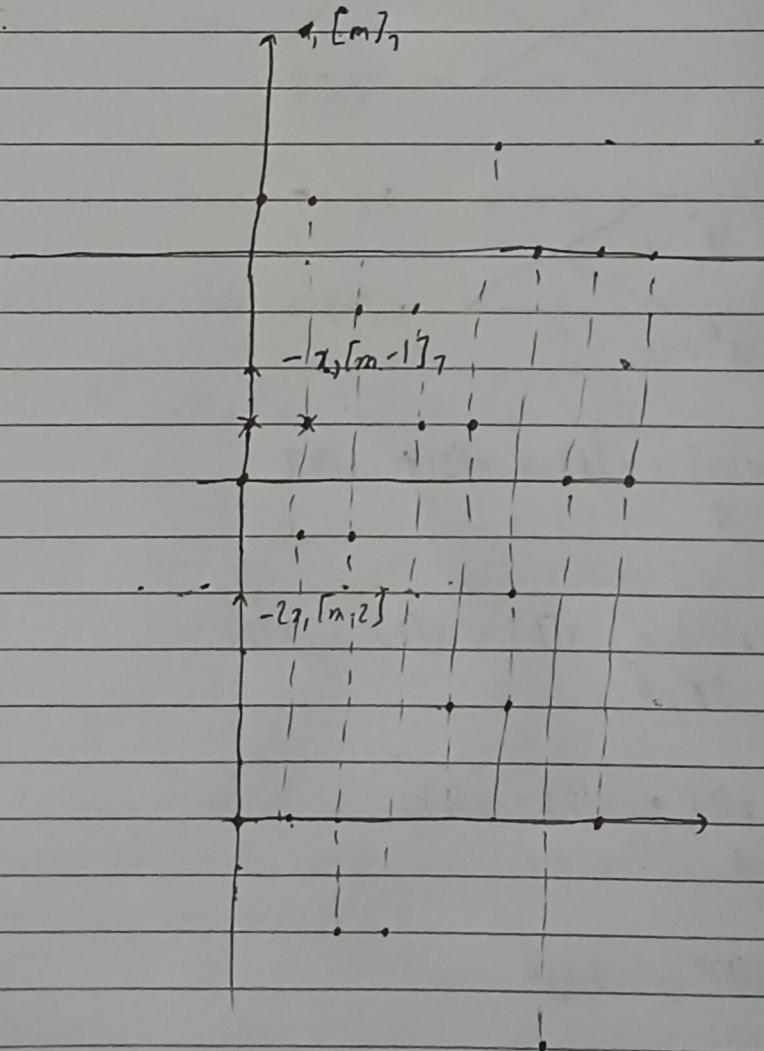
$$= x_1[0]x_2[m] + x_1[1]x_2[m-1] + x_1[2]x_2[m-2] + \dots + x_1[7]x_2[m-7]$$

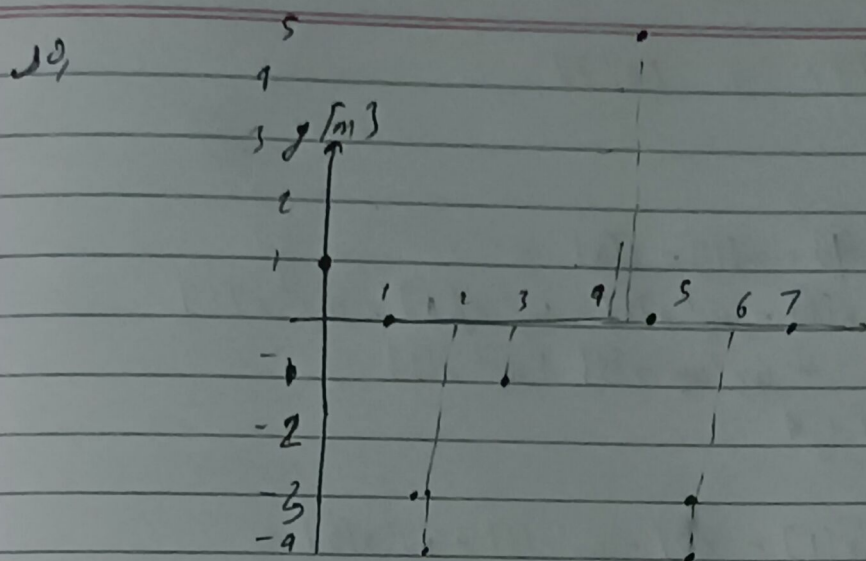
$$= \sum_{n=0}^7 x_1[n] x_2[m-n]$$

$$= x_1[0]x_2[m] + x_1[1]x_2[m-1] + x_1[2]x_2[m-2]$$

$$= x_1[m] - x_2[m] - 2x_1[m-2]$$

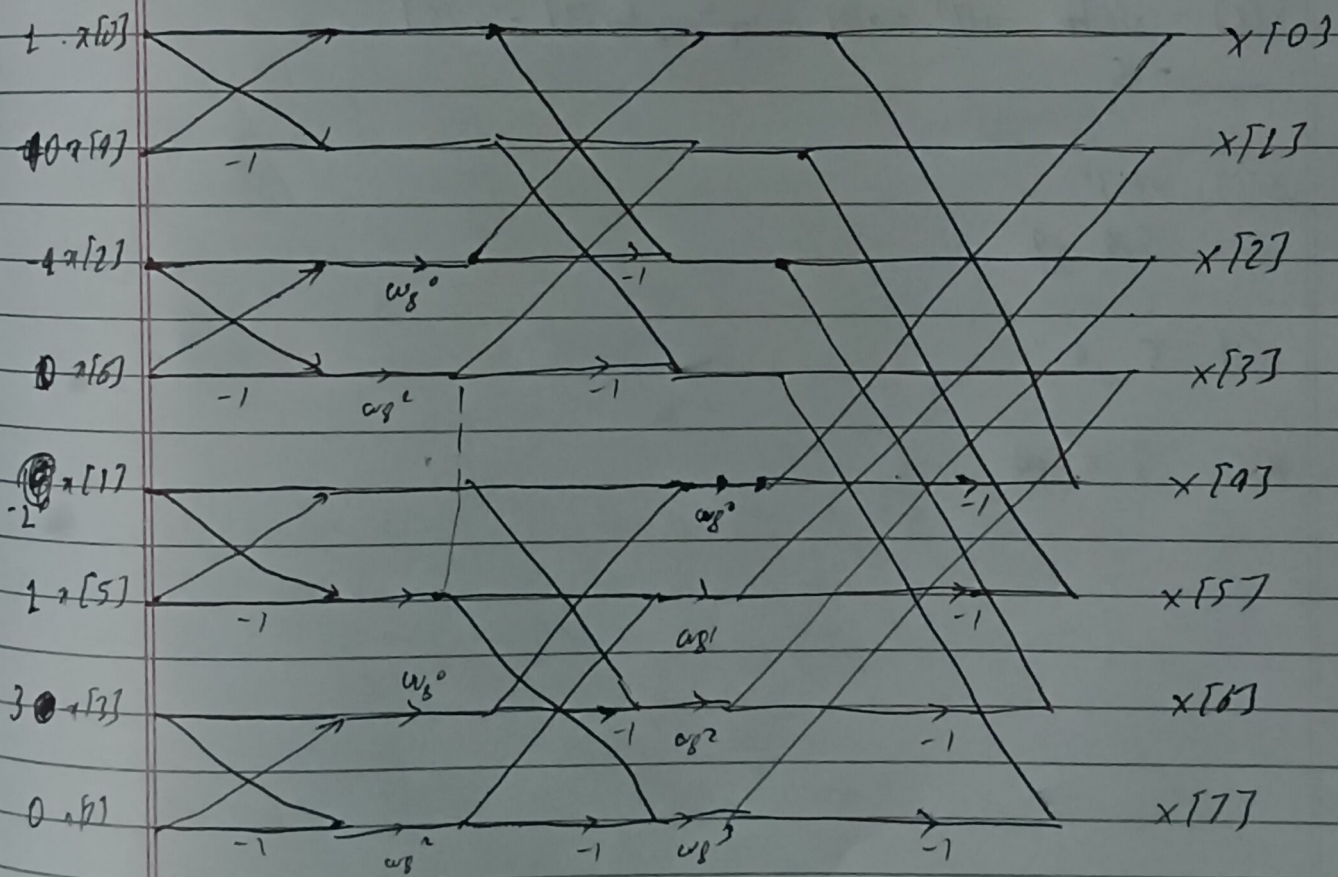
graphically:





$y[m] = \{1, 0, -4, -1, 5, 0, -4, 0\}$

A
SLBH
DTT FFT



$$X[0] = x[0] + x[4] + x[8] + x[12] \\ = -1$$

$$X[1] = x[0] - x[4] + x[8] - x[12] \\ = x[0] - x[4] + w_8^2 x[8] - w_8^2 x[12] - w_8^4 x[5] \\ + w_8^6 w_8^4 x[3] + w_8^4 x[1] \\ = -3 - 24 + j4$$

$$X[2] = x[0] + w_8^4 x[4] - x[8] - w_8^2 x[12] + w_8^2 x[5] \\ = 5 + j4$$

$$X[3] = x[0] + w_8^3 x[4] - w_8^6 x[8] - w_8^4 w_8^3 x[12] - w_8^3 x[5] \\ = 5 - 24 - j4$$

$$X[4] = x[0] - x[4] + x[8] - w_8^2 w_8^4 x[12] - x[5] \\ = -5$$

$$X[5] = x[3]^* \\ = 5 - 24 + j4$$

$$X[6] = 5 - j4$$

$$X[7] = -3 - 24 - j4$$

70th

Prove circular convolution.

statement:

$$x_1[n] \xleftrightarrow{\text{DFT}} X_1[k]$$

$$x_2[n] \xleftrightarrow{\text{DFT}} X_2[k]$$

$$x_1[n] \otimes x_2[n] \xleftrightarrow{\text{DFT}} X_1[k] X_2[k]$$

Here,

$$X_1[k] = \sum_{n=0}^{N-1} x_1[n] e^{-j \frac{2\pi k}{N} n}$$

$$X_2[k] = \sum_{n=0}^{N-1} x_2[n] e^{-j \frac{2\pi k}{N} n}$$

$$\text{let } x_3[k] = x_1[k] \cdot x_2[k]$$

too complicated

of G&T method

new method
after this

IDFT of $x_3[k]$ is $x_3[n]$ given as:

$$x_3[n] = \frac{1}{N} \sum_{k=0}^{N-1} x_3[k] e^{j \frac{2\pi k}{N} n}$$

$$= \frac{1}{N} \sum_{k=0}^{N-1} x_1[k] x_2[k] e^{j \frac{2\pi k}{N} n}$$

$$= \frac{1}{N} \sum_{k=0}^{N-1} \left[\sum_{l=0}^{N-1} x_1[l] e^{-j \frac{2\pi k}{N} l} \right] \left[\sum_{m=0}^{N-1} x_2[m] e^{-j \frac{2\pi k}{N} m} \right] e^{j \frac{2\pi k}{N} n}$$

$$= \frac{1}{N} \sum_{l=0}^{N-1} x_1[l] \sum_{m=0}^{N-1} x_2[m] \sum_{k=0}^{N-1} e^{j \frac{2\pi k}{N} (n-l-m)}$$

in the form of:

$$\sum_{k=0}^{N-1} a^k = \begin{cases} N & \text{for } a = 1 \\ \frac{1-a^N}{1-a} & \text{for } a \neq 1 \end{cases}$$

$$a = e^{j \frac{2\pi k}{N} (n-l-m)}$$

$a = L$ if $m - n - L = N$, or $a^N = L$, for $a \neq 0$.

so,

$$\sum_{k=0}^{N-1} a_k = \begin{cases} N & \text{for } L = m - n + PN = [m - n]_N, \text{ } P \text{ is an integer} \\ & \text{otherwise} \end{cases} \quad \& \text{ } L = 0 \text{ to } N-1$$

so,

$$x_3[m] = \sum_{n=0}^{N-1} x_1[n] \sum_{l=0}^{N-1} x_2[l] \cdot \frac{1}{N} \times N$$

$$\text{DFT} [x_1[n] \otimes x_2[n]] = \sum_{n=0}^{N-1} (x_1[n] \otimes x_2[n]) e^{-j \frac{2\pi k}{N} n}$$

$$= \sum_{n=0}^{N-1} \left(\sum_{m=0}^{N-1} x_1[m] x_2[n-m]_N \right) e^{-j \frac{2\pi k}{N} n}$$

$$= \sum_{m=0}^{N-1} x_1[m] \sum_{n=0}^{N-1} x_2[n-m]_N e^{-j \frac{2\pi k}{N} n}$$

let $n - m = p$

$n = m + p$, limit changes to $p = -m$ to $p = -m + N - 1$

since periodic, we rewrite limits as 0 to $N-1$

then,

$$= \sum_{m=0}^{N-1} x_1[m] \sum_{p=0}^{N-1} x_2[p] e^{-j \frac{2\pi k}{N} (m+p)}$$

$$= \sum_{m=0}^{N-1} x_1[m] e^{-j \frac{2\pi k}{N} m} \sum_{p=0}^{N-1} x_2[p] e^{-j \frac{2\pi k}{N} p}$$

↓

$$= X_L[k] \times X_L[k] \quad \text{// proved}$$